# Benjamin DUPONT

Zurich, Switzerland | dupontben@proton.me | linkedin.com/in/ben-dupont | ben-du-pont.github.io

## Summary

Robotics software engineer at Ascento in Zurich. MSc in Mechanical Engineering (robotics) from ETH Zurich, with experience at NASA JPL, Formula 1, and Formula Student. Proficient in Python, C++, MATLAB, and ROS for end-to-end robotics pipelines, including perception, state estimation, control, simulation, and validation.

## Experience

**Robotics Software Engineer** Oct 2025 – Present
*Ascento AG* *Zurich, Switzerland*

- Localization and autonomy software (SLAM, sensor fusion) for autonomous outdoor security robots

**Autonomy and Perception Research Intern** Jan 2025 – Aug 2025
*NASA Jet Propulsion Laboratory - Aerial Mobility Group* *Pasadena, CA, USA*

- Developed complete autonomy software stack for Mars rotorcraft platform, including mission planning, execution systems, and failsafe mechanisms
- Implemented terrain mapping and landing site detection algorithms for autonomous safe landing in challenging environments
- Built Ground Control Station with long-range telemetry capabilities and conducted extensive simulation and hardware-in-the-loop testing
- Integrated visual-inertial navigation systems with sensor fusion and validated performance through field testing

**Head of Controls** Aug 2023 – March 2024
*AMZ Racing* *Zurich, Switzerland*

- Recruited and led a controls team of 5 engineers, going on to win multiple European Formula Student competitions
- Successfully orchestrated migration from ROS1 to ROS2 while implementing new control algorithms in ambitious time constraints
- Established structured code review processes ensuring high-quality implementation across the controls subsystem

**Controls Engineer** Sept 2022 – Aug 2023
*AMZ Racing* *Zurich, Switzerland*

- Developed MPC that enabled autonomous racing at speeds up to 16 m/s (58 km/h), a significant improvement from initial testing
- Implemented delay compensation techniques that improved trajectory tracking and stability on cone-delimited tracks
- Successfully bridged sim-to-real gap by adapting controllers to accommodate real-world vehicle dynamics differences

**Teaching Assistant - Robotics** June 2024 – Dec 2024
*ETH Zurich - Self-Driving Cars with Duckietown* *Zurich, Switzerland*

- Designed comprehensive exercises on path planning and state estimation that strengthened students' practical skills
- Mentored student groups through challenging projects while providing technical hardware troubleshooting support

## Education

**Eidgenoessische Technische Hochschule (ETH)** Zurich, Switzerland
*Master of Science (MSc), Mechanical Engineering* *2022 – 2025*

- Major in robotics (controls, computer vision, state estimation, ML/RL); Minor in fluid dynamics
- Confirmed average grade of 5.45/6

**Ecole Polytechnique Federale (EPFL)** Lausanne, Switzerland
*Bachelor of Science (BSc), Mechanical Engineering* *2019 – 2022*

- Graduated with top-tier average grade of 5.01/6; Exchange at NTNU (Norway, 2021-2022)

## Technical Skills

**Programming**: Python, MATLAB/Simulink, C, C++, ROS/ROS2, Linux, Docker, Bash scripting, Gazebo, PX4 Autopilot

**AI & Robotics**: Computer vision (OpenCV), deep learning (ML/RL), state estimation, path planning, control theory

**Drone Systems**: FPV drone building and piloting, flight controller configuration, hardware troubleshooting, real-world flight operations

**Languages**: English and French (bilingual), German (B1–B2), Spanish (A1)

## PROJECTS

• **Full Monocular Visual Odometry Pipeline**: Implemented complete visual odometry system using two-view geometry, KLT tracking, P3P RANSAC, and bundle adjustment to reconstruct 3D landmarks and achieve high local trajectory consistency even during complex camera motions, across multiple datasets

• **Duckietown Intersection Navigation**: Developed computer vision-based algorithm enabling autonomous robots to recognize intersections, stop at lines, and interact with other robots by interpreting indicators and yielding right of way

• **Reinforcement Learning for Racing**: Created vehicle dynamics simulator with Pacejka tire model for Formula Student-like tracks, currently implementing PPO model to transition from traditional PID control to reinforcement learning approaches